

1Voucher — Technical Docs

Operator reference for the platform's value model, contracts, functions and the on-chain events that drive reporting. For a high-level product overview, see the **About** page.

Concepts Payments Gas fees Merchant Contracts Functions Events & Reporting

API

Gas fees — who pays, and how

Consumers never pay or see gas. The **platform sponsors every transaction**. Today that sponsorship works through a **relay** — the backend submits the user's signed transaction and pays the gas directly. This is deliberately simpler than full ERC-4337: there is no bundler or paymaster in the path yet. At scale it should move to a bundler + paymaster (Pimlico); the reasons are below.

How it works today — relayed Safe transactions (not quite ERC-4337)

1. The consumer's passkey signs the exact **SafeTx** (recipient, amount, currency, nonce) for `Vault.transfer` — a biometric prompt, no gas.
2. The backend submits an **ordinary Ethereum transaction** calling `Safe.execTransaction(...)`, carrying that signature in the `signatures` argument.
3. Because it's an ordinary tx, **the sender pays the gas in ETH**. The SafeTx uses `gasPrice = 0`, so the Safe does *not* reimburse the sender — the backend simply absorbs the cost.

The relay is the platform's backend hot wallet (`BACKEND_SIGNER`). It can relay or stall a transfer but **cannot forge or alter it** — the passkey already signed the exact parameters, so the backend is a transport, not a custodian. Self-custody is preserved.

Why "not quite ERC-4337": there is no EntryPoint, no bundler, no paymaster and no `UserOperation` . The wallet is a Safe 1.4.1 smart account (threshold 1, passkey owner) driven by its normal `execTransaction` path. Newly-registered Safes are 4337-capable (the Safe4337Module is enabled at setup), but submission still goes through the relay — so the door to a paymaster is open without re-onboarding anyone.

Action	Who submits	Who pays gas	Consumer pays?
P2P transfer (Vault.transfer)	Backend relay	Backend hot wallet (ETH)	No
Registration, KYC update, harvest, merchant whitelist	Backend	Backend hot wallet (ETH)	No
Contract upgrades	Owner / governance wallet	Owner wallet (ETH)	—